Chapter 24

Multivariable Calculus for Probability and Statistics

Densities, expectations, transformations, likelihood, and local approximation

Probability and statistics give one of the most important modern uses of multivariable calculus. A probability model is often a function on a space of possible outcomes. A statistical model is often a function on a space of unknown parameters. In both cases, we need the language of multivariable calculus: domains, level curves, multiple integrals, gradients, Hessians, Jacobians, and change of variables.

This chapter is a bridge from the core calculus of several variables to later courses in probability, mathematical statistics, data science, and machine learning. The central idea is simple.

Central idea

A density is a function whose integral gives probability. A likelihood is a function whose maximum gives an estimate.

The first sentence uses multiple integrals. The second uses gradients and optimization.

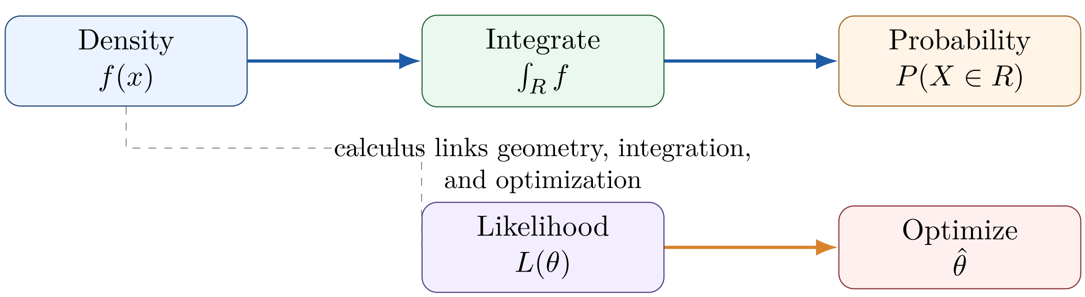

Figure 24.1: Two central roles of multivariable calculus in statistics: integration over outcome space and optimization over parameter space.

Learning goals

After studying this chapter, you should be able to:

1. interpret a joint density $f_{X,Y}(x, y)$ geometrically and probabilistically;
2. compute probabilities and expectations using double and triple integrals;
3. find marginal and conditional densities from a joint density;
4. use change of variables and Jacobians to transform random variables;
5. interpret covariance matrices through ellipses and quadratic forms;
6. use gradients and Hessians to understand likelihood functions;
7. connect Taylor approximation to standard errors and uncertainty quantification;
8. use numerical integration and simulation to check analytic calculations.

Throughout the chapter, keep two pictures in mind. A joint density is like a surface whose total volume is 1. A log-likelihood is like a landscape whose highest point gives the best-fitting parameter.

24.1 Random variables as functions and data as points

A random variable is a function from a sample space to numbers. A random vector is a function from a sample space to a vector space.

For example, if we observe height and weight from a randomly selected person, then the outcome can be written as a vector

$$X = (X_1, X_2) = (\text{height}, \text{weight}) \in \mathbb{R}^2.$$

If we observe n measurements, the dataset can be written as a point in $\mathbb{R}^n$:

$$x = (x_1, x_2, \ldots, x_n).$$

If we fit a model with parameters $\theta_1, \ldots, \theta_p$, the parameter vector is a point in parameter space:

$$\theta = (\theta_1, \ldots, \theta_p) \in \mathbb{R}^p.$$

This is why multivariable calculus appears everywhere in statistics. We integrate over outcome spaces, differentiate over parameter spaces, and use linear and quadratic approximations to measure uncertainty.

Three spaces in statistics

A statistical problem often involves three different spaces:

- **Sample space:** where random outcomes live.
- **Data space:** where observed datasets live.
- **Parameter space:** where unknown model parameters live.

Multivariable calculus helps us move among all three.

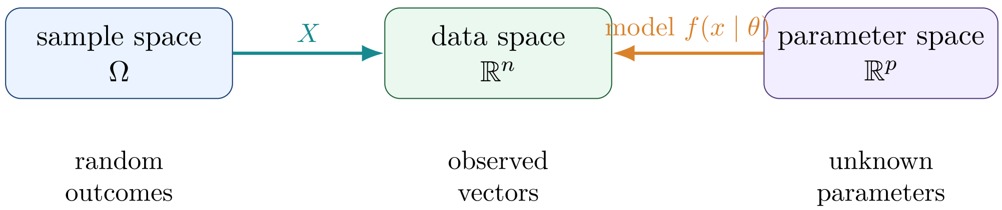

Figure 24.2: Statistics moves among sample space, data space, and parameter space. Multivariable calculus supplies the geometry and computation.

24.2 Joint densities and probability as volume

A continuous random vector (X, Y) has a joint density $f_{X,Y}(x, y)$ if probabilities are computed by double integrals:

$$\mathbb{P}((X, Y) \in R) = \iint_R f_{X,Y}(x, y)\, dA.$$

For $f_{X,Y}$ to be a valid density, it must satisfy

$$f_{X,Y}(x, y) \geq 0, \qquad \iint_{\mathbb{R}^2} f_{X,Y}(x, y)\, dA = 1.$$

The probability of a region is the volume under the density surface above that region.

Example 24.1: A simple density on a square

Let

$$f(x, y) = x + y$$

on the unit square $0 \leq x \leq 1$, $0 \leq y \leq 1$, and let $f(x, y) = 0$ outside the square. This is a probability density because

$$\int_0^1 \int_0^1 (x + y)\, dy\, dx = \int_0^1 \left(x + \frac{1}{2}\right) dx = 1.$$

Now compute $\mathbb{P}(X + Y \leq 1)$. The region is the triangle $0 \leq x \leq 1$, $0 \leq y \leq 1 - x$. Thus

$$\mathbb{P}(X + Y \leq 1) = \int_0^1 \int_0^{1-x} (x + y)\, dy\, dx.$$

The inner integral is

$$\int_0^{1-x} (x + y)\, dy = x(1 - x) + \frac{(1 - x)^2}{2}.$$

Therefore,

$$\mathbb{P}(X + Y \leq 1) = \int_0^1 \left[x(1 - x) + \frac{(1 - x)^2}{2}\right] dx = \frac{1}{2}.$$

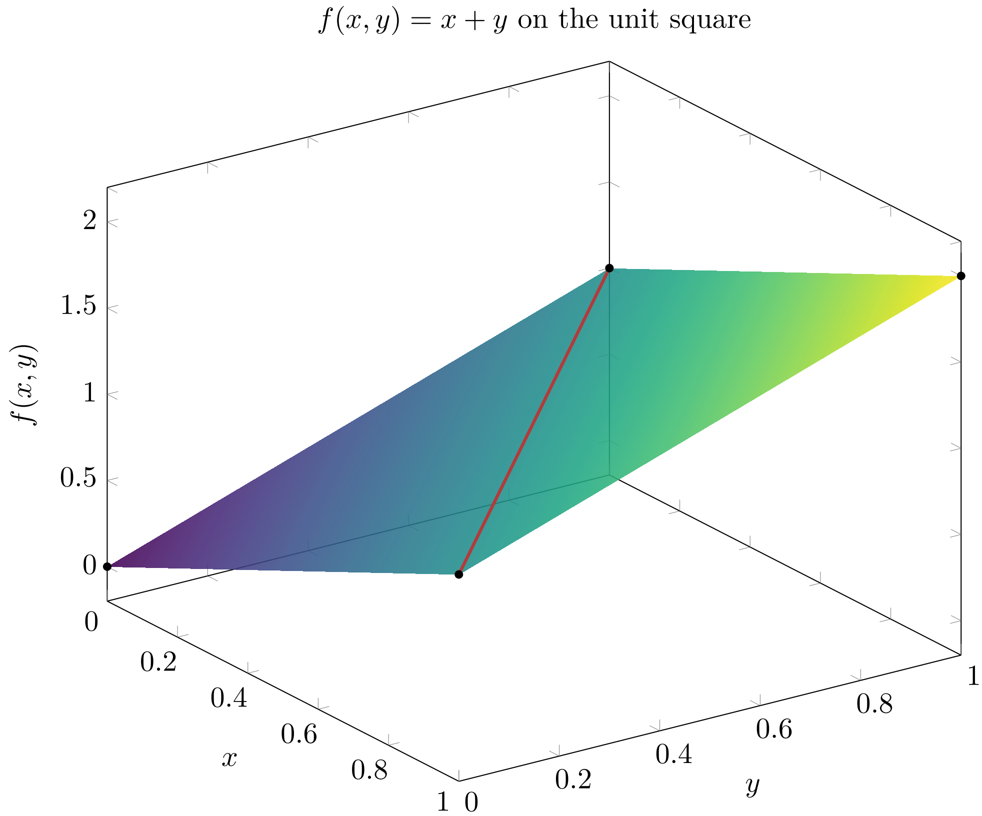

Figure 24.3: The density $f(x,y) = x + y$ on the unit square. Probability over a region is volume under the surface over that region.

24.3 Marginal densities

A joint density describes the simultaneous behavior of two random variables. A marginal density describes one variable alone. If (X,Y) has joint density $f_{X,Y}$, then the marginal density of X is

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dy.$$

Similarly,

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dx.$$

The operation is geometric: we collapse the density surface in one direction by integrating out the other variable.

Example 24.2: Marginals for $f(x,y) = x + y$

On the unit square,

$$f_X(x) = \int_0^1 (x+y)\,dy = x + \frac{1}{2}.$$

Similarly,

$$f_Y(y) = \int_0^1 (x+y)\,dx = y + \frac{1}{2}.$$

Check normalization:

$$\int_0^1 f_X(x)\,dx = \int_0^1 \left(x + \frac{1}{2}\right)\,dx = 1.$$

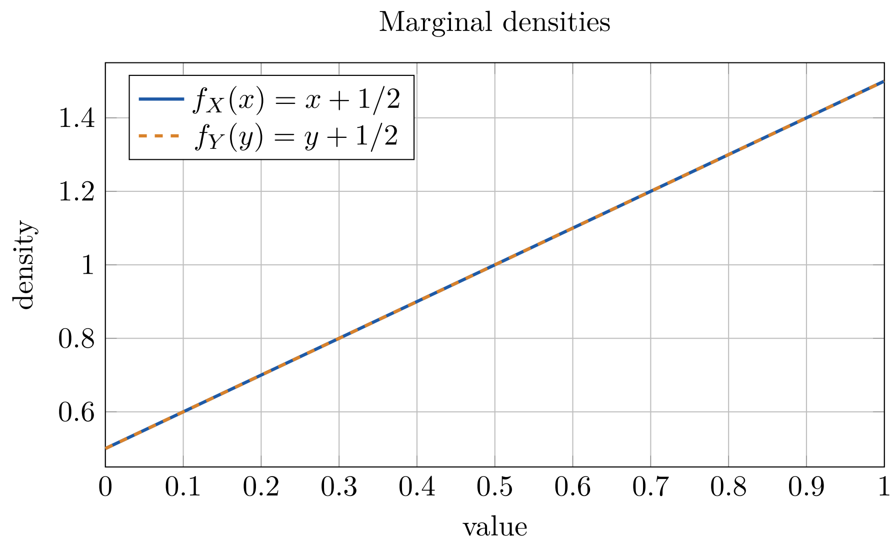

Figure 24.4: Marginal densities are obtained by integrating the joint density in one variable.

Integrating out variables

In statistics and machine learning, the phrase **integrate out** means to remove a variable by integration. Marginalization is integration.

24.4 Conditional densities and independence

The conditional density of Y given $X = x$ is

$$f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x, y)}{f_X(x)},$$

provided $f_X(x) > 0$. This is the continuous analogue of conditional probability:

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Example 24.3: Conditional density from a joint density

For $f(x, y) = x + y$ on the unit square,

$$f_{Y|X}(y \mid x) = \frac{x + y}{x + 1/2}, \qquad 0 \le y \le 1.$$

For fixed x, this is a density in y:

$$\int_0^1 \frac{x+y}{x+1/2}\,dy = \frac{x+1/2}{x+1/2} = 1.$$

The conditional mean of Y given $X = x$ is

$$\mathbb{E}[Y \mid X = x] = \int_0^1 y\,\frac{x+y}{x+1/2}\,dy = \frac{x/2+1/3}{x+1/2}.$$

This conditional mean depends on x, so knowing X changes the distribution of Y.

Random variables X and Y are independent if

$$f_{X,Y}(x,y) = f_X(x)f_Y(y)$$

for all relevant (x,y). For $f(x,y) = x+y$, we have

$$f_X(x)f_Y(y) = \left(x+\frac{1}{2}\right)\left(y+\frac{1}{2}\right),$$

which is not equal to $x+y$. Therefore, X and Y are not independent.

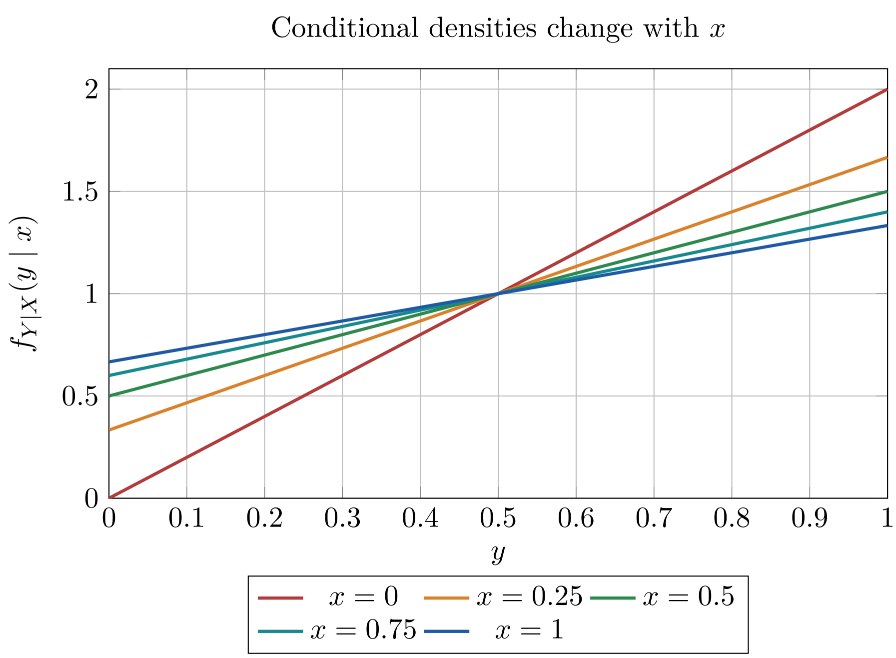

Figure 24.5: Conditional densities $f_{Y|X}(y \mid x)$ for several fixed values of x.

24.5 Expectation as integration

For a function $g(X,Y)$, the expected value is

$$\mathbb{E}[g(X,Y)] = \iint_{\mathbb{R}^2} g(x,y)f_{X,Y}(x,y)\,dA.$$

Important special cases include

$$\mathbb{E}[X] = \iint x f_{X,Y}(x,y)\,dA, \qquad \mathbb{E}[Y] = \iint y f_{X,Y}(x,y)\,dA,$$

$$\mathbb{E}[XY] = \iint xy f_{X,Y}(x,y)\,dA.$$

The covariance is

$$\mathrm{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

The covariance matrix of a random vector $X = (X_1, \ldots, X_n)$ is

$$\Sigma = \mathbb{E}[(X-\mu)(X-\mu)^T].$$

It is a matrix version of variance.

Example 24.4: Means and covariance for a simple joint density

For $f(x,y) = x + y$ on the unit square,

$$\mathbb{E}[X] = \int_0^1 \int_0^1 x(x+y)\,dy\,dx = \frac{7}{12},$$

and by symmetry,

$$\mathbb{E}[Y] = \frac{7}{12}.$$

Also,

$$\mathbb{E}[XY] = \int_0^1 \int_0^1 xy(x+y)\,dy\,dx = \frac{1}{3}.$$

Thus

$$\mathrm{Cov}(X,Y) = \frac{1}{3} - \frac{7}{12} \cdot \frac{7}{12} = -\frac{1}{144}.$$

The covariance is slightly negative. This is a small but real dependence.

24.6 Numerical integration on grids

Many integrals in probability and statistics do not have closed-form antiderivatives. A practical alternative is numerical integration. Suppose $f(x,y)$ is evaluated on a rectangular grid. Then

$$\iint_R f(x,y)\,dA \approx \sum_{i,j} f(x_i, y_j)\Delta x \Delta y.$$

This is the same Riemann-sum idea from multiple integration, now used for probabilities and expectations.

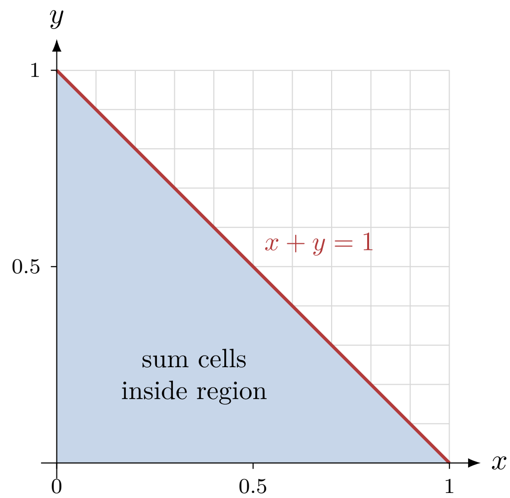

Figure 24.6: Numerical integration estimates the probability of a region by summing density values over grid cells. The exact answer in Example 24.1 is 1/2.

24.7 Change of variables for random variables

Change of variables is one of the most important applications of the Jacobian. Suppose (U, V) has density $f_{U,V}(u, v)$, and suppose

$$(x, y) = T(u, v).$$

If T is one-to-one on the relevant region and has nonzero Jacobian determinant, then the density of $(X, Y) = T(U, V)$ is

$$f_{X,Y}(x, y) = f_{U,V}(u(x, y), v(x, y)) \left| \det \frac{\partial(u, v)}{\partial(x, y)} \right|.$$

Equivalently, if we integrate in (u, v)-coordinates, then

$$dx\, dy = \left| \det \frac{\partial(x, y)}{\partial(u, v)} \right| du\, dv.$$

The Jacobian measures local area distortion.

Example 24.5: Uniform distribution on a disk

A common mistake is to sample a radius uniformly when trying to sample uniformly from a disk. If R is uniform on $[0, 1]$ and Θ is uniform on $[0, 2\pi]$, then points concentrate near the center. For uniform sampling in the unit disk, use

$$R = \sqrt{U}, \qquad \Theta = 2\pi V,$$

where U, V are independent uniform random variables on $[0, 1]$. Then

$$X = R \cos \Theta, \qquad Y = R \sin \Theta.$$

Why the square root? Because the area of a disk of radius r is proportional to r^2, so

$$\mathbb{P}(R \le r) = r^2.$$

Thus R^2 should be uniform on $[0, 1]$.

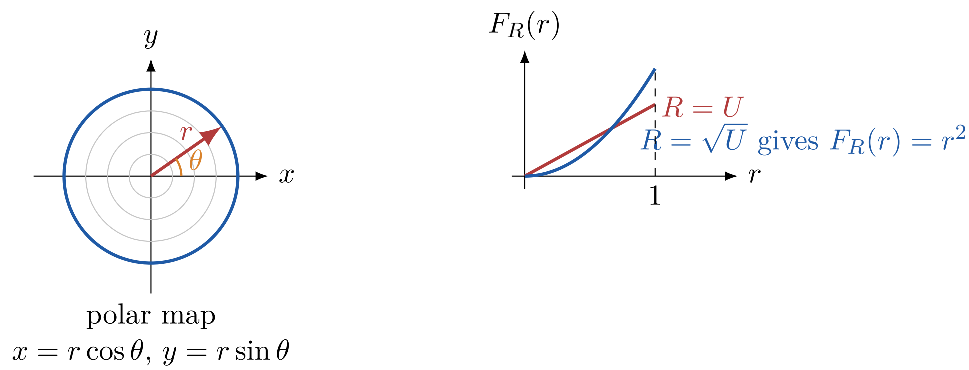

Figure 24.7: Uniform area in a disk requires the radial distribution $\mathbb{P}(R \le r) = r^2$, hence $R = \sqrt{U}$.

24.8 The bivariate normal distribution

The multivariate normal distribution is the most important continuous distribution in multivariable statistics. A random vector $X \in \mathbb{R}^n$ has multivariate normal density with mean vector μ and covariance matrix Σ if

$$f(x) = \frac{1}{(2\pi)^{n/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)\right).$$

For $n = 2$, the level curves of the density are ellipses:

$$(x-\mu)^T\Sigma^{-1}(x-\mu) = c.$$

This is a quadratic form. The eigenvectors of Σ determine the principal directions of the ellipses, and the eigenvalues determine their squared spread in those directions.

Geometry of covariance

The covariance matrix is not just a table of numbers. It is a geometric object. It determines the directions and scales of uncertainty.

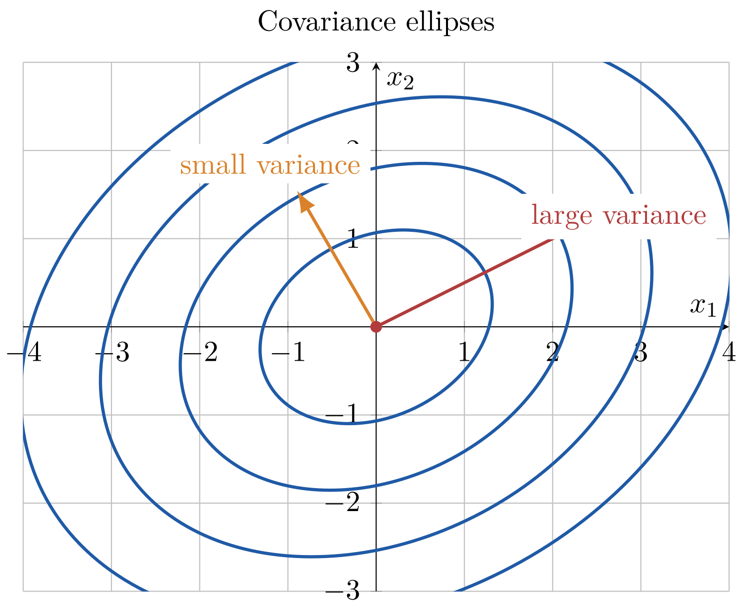

Figure 24.8: Level curves of a bivariate normal density are ellipses. The covariance matrix controls the directions and scales.

24.9 Likelihood functions as multivariable functions

In statistics, the unknown parameter is often a vector

$$\theta = (\theta_1, \ldots, \theta_p).$$

Given data $x_1, \ldots, x_n$, the likelihood function is

$$L(\theta) = f(x_1, \ldots, x_n \mid \theta).$$

The log-likelihood is

$$\ell(\theta) = \log L(\theta).$$

Maximum likelihood estimation asks us to find

$$\hat{\theta} = \operatorname*{argmax}_{\theta} \ell(\theta).$$

This is a multivariable optimization problem.

Example 24.6: Normal model with unknown mean and variance

Suppose $x_1, \ldots, x_n$ are modeled as independent normal observations with mean μ and standard deviation $\sigma > 0$. The log-likelihood is, up to an additive constant,

$$\ell(\mu, \sigma) = -n \log \sigma - \frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i - \mu)^2.$$

The maximum occurs at

$$\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \hat{\mu})^2.$$

This is a two-variable optimization problem on the half-plane $\sigma > 0$.

A normal log-likelihood landscape

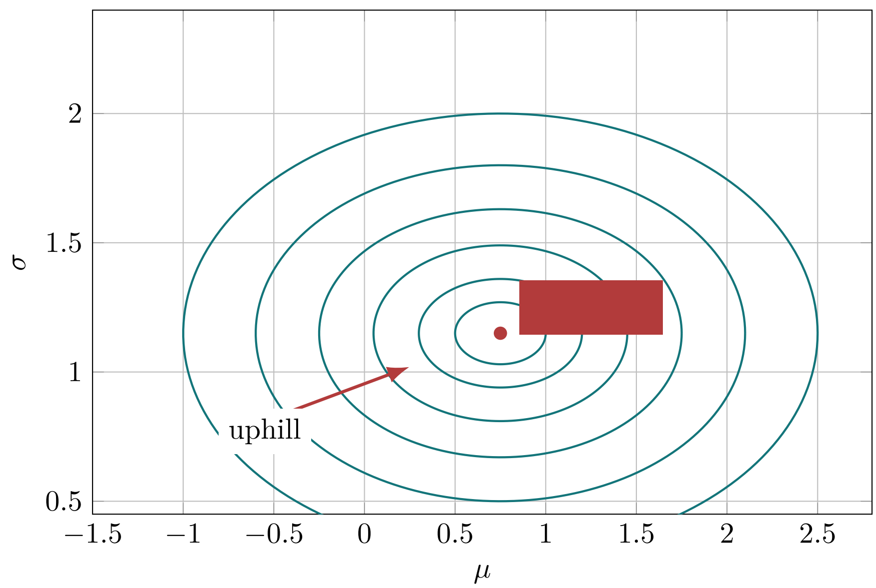

Figure 24.9: A log-likelihood is a function on parameter space. Its maximum gives the maximum likelihood estimate.

24.10 Score, Hessian, and information

The gradient of the log-likelihood is called the **score**:

$$S(\theta) = \nabla \ell(\theta).$$

At an interior maximum likelihood estimate, the score is zero:

$$\nabla \ell(\hat{\theta}) = 0.$$

The Hessian matrix is

$$H(\theta) = \nabla^2 \ell(\theta).$$

At a local maximum, the Hessian is usually negative definite. The negative Hessian

$$-\nabla^2 \ell(\hat{\theta})$$

measures local curvature of the log-likelihood near the maximum. In many statistical models, this curvature is connected to information and standard errors.

A second-order Taylor approximation gives

$$\ell(\theta) \approx \ell(\hat{\theta}) + \frac{1}{2}(\theta - \hat{\theta})^T \nabla^2 \ell(\hat{\theta})(\theta - \hat{\theta}).$$

Since the Hessian is negative definite near a maximum, the log-likelihood looks locally like an upside-down quadratic bowl.

Curvature and uncertainty

A sharply curved log-likelihood means the parameter is estimated precisely. A flat log-likelihood means the data do not strongly identify the parameter.

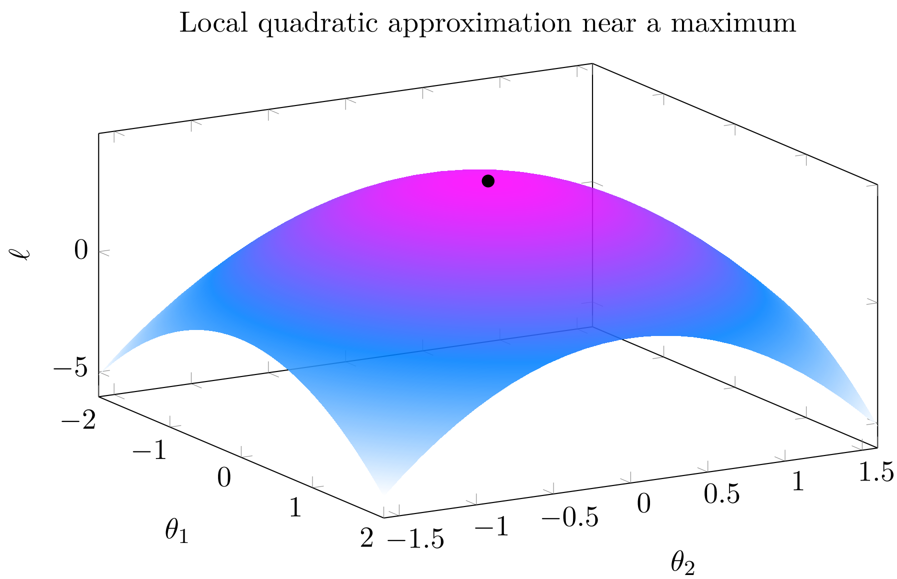

Figure 24.10: Near an interior maximum, the second-order Taylor approximation makes the log-likelihood look like an upside-down quadratic bowl.

24.11 The delta method as linear approximation

Suppose an estimator $\hat{\theta}$ is approximately normal:

$$\hat{\theta} \approx N(\theta, \Sigma).$$

If we want to understand $g(\hat{\theta})$ for a differentiable function g, we use linear approximation:

$$g(\hat{\theta}) \approx g(\theta) + \nabla g(\theta)^T (\hat{\theta} - \theta).$$

Therefore,

$$\mathrm{Var}(g(\hat{\theta})) \approx \nabla g(\theta)^T \Sigma \nabla g(\theta).$$

This is the multivariable delta method. It is just the linearization from Chapter 9, applied to statistical uncertainty.

Example 24.7: Ratio of two estimates

Suppose $\hat{\theta} = (\hat{a}, \hat{b})$ has covariance matrix Σ, and we want the uncertainty of

$$g(a, b) = \frac{a}{b}.$$

The gradient is

$$\nabla g(a, b) = \left(\frac{1}{b}, -\frac{a}{b^2}\right).$$

So

$$\operatorname{Var}\left(\frac{\hat{a}}{\hat{b}}\right) \approx \begin{bmatrix} 1/b & -a/b^2 \end{bmatrix} \Sigma \begin{bmatrix} 1/b \\ -a/b^2 \end{bmatrix}.$$

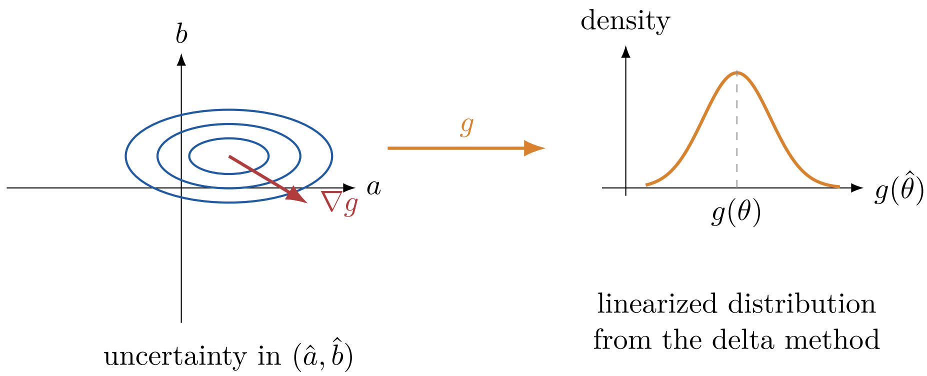

Figure 24.11: The delta method approximates the distribution of a nonlinear transformation by linearizing the transformation near the mean.

24.12 Monte Carlo integration

Monte Carlo integration uses random samples to estimate integrals. If $U_1, \ldots, U_N$ are uniformly sampled from a region R with area $A(R)$, then

$$\iint_R f(x, y)\, dA \approx A(R) \frac{1}{N} \sum_{i=1}^{N} f(U_i).$$

If $X_1, \ldots, X_N$ are sampled from a density $p(x)$, then

$$\mathbb{E}[g(X)] \approx \frac{1}{N} \sum_{i=1}^{N} g(X_i).$$

This is one of the key computational ideas behind simulation-based statistics, Bayesian computation, and machine learning.

Example 24.8: Estimating an integral by simulation

Estimate

$$I = \int_0^1 \int_0^1 e^{-(x^2+y^2)}\, dy\, dx.$$

If (U, V) is uniform on the unit square, then

$$I = \mathbb{E}[e^{-(U^2+V^2)}].$$

So we can approximate I by a sample average.

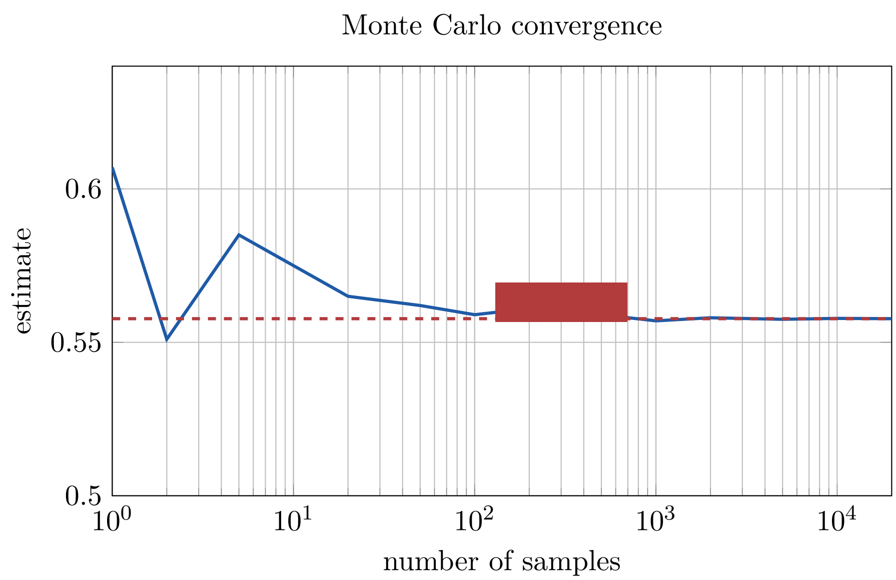

Figure 24.12: Monte Carlo integration estimates an integral by averaging random function values. The estimate stabilizes as the sample size grows.

24.13 Bayesian posterior as a normalized density

Bayesian statistics combines a likelihood and a prior distribution. If θ is a parameter and x is observed data, Bayes' rule gives

$$p(\theta \mid x) = \frac{p(x \mid \theta)p(\theta)}{p(x)}.$$

The denominator

$$p(x) = \int p(x \mid \theta)p(\theta)\, d\theta$$

is a normalizing constant. In multivariable problems,

$$p(x) = \int_\Theta p(x \mid \theta)p(\theta)\, d\theta$$

may be a high-dimensional integral. This is why numerical integration, Monte Carlo methods, and optimization are so important in modern Bayesian statistics.

Example 24.9: A two-parameter posterior landscape

Suppose a posterior density for two parameters is proportional to

$$\exp\left[-\frac{1}{2}(\theta-\mu)^T A(\theta-\mu)\right].$$

The contours are ellipses. The matrix A is a precision matrix, and A^{-1} is the covariance matrix.

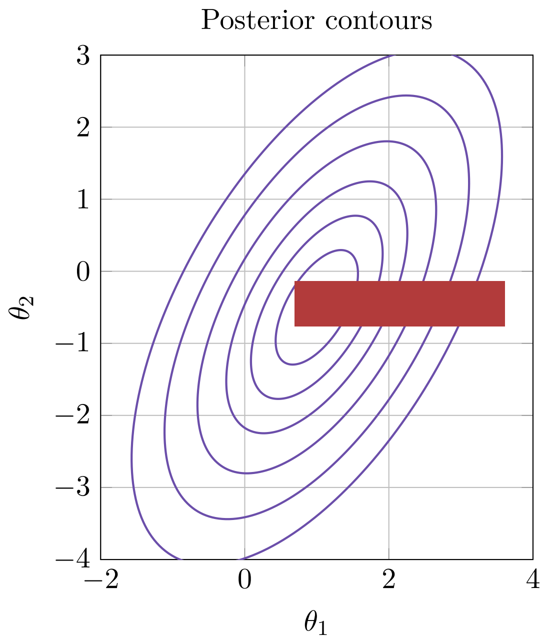

Figure 24.13: A two-parameter posterior density. The highest point is the posterior mode, and the contour shape describes uncertainty.

24.14 What to remember

Multivariable calculus gives statistics its geometry and computation.

- A joint density is a nonnegative function with total integral 1.
- A probability is an integral over a region.
- A marginal density is obtained by integrating out variables.
- A conditional density is a normalized slice of the joint density.
- An expectation is an integral of a function against a density.
- A covariance matrix is a geometric object describing spread and direction.
- A Jacobian accounts for local volume distortion under a change of variables.
- A likelihood is a function on parameter space.
- A score is the gradient of a log-likelihood.

- A Hessian describes local curvature and uncertainty.
- Monte Carlo methods approximate integrals by random averages.

24.15 Python lab

In the accompanying lab, you will:

1. plot joint densities as surfaces and contour maps;
2. compute marginal and conditional densities numerically;
3. simulate points from a disk using the correct radial transformation;
4. visualize covariance ellipses of a bivariate normal distribution;
5. plot a normal log-likelihood surface for unknown mean and variance;
6. approximate a nonlinear estimator using the delta method;
7. estimate probabilities and expectations using Monte Carlo simulation.

Chapter resources

- Lab notebook: `../labs/chapter-24-multivariable-calculus-for-probability-and-statistics.ipynb`
- Interactive page: `../htmls/chapter-24-multivariable-calculus-for-probability-and-statistics.html`

24.16 Exercises

Conceptual questions

1. Explain why a joint density must integrate to 1.
2. In your own words, explain the difference between a marginal density and a conditional density.
3. Why is a covariance matrix a geometric object?
4. Why does a change of variables formula need a Jacobian factor?
5. What is the difference between likelihood and probability?
6. Explain why the gradient of the log-likelihood is zero at an interior maximum likelihood estimate.
7. Explain how the Hessian of the log-likelihood is related to uncertainty.
8. What does Monte Carlo integration estimate, and why does it work?

Computation and derivation

1. Let $f(x, y) = 2$ on the triangle $0 \le y \le x \le 1$ and 0 elsewhere. Verify that f is a joint density.

2. For the density in Problem 1, compute $f_X(x)$ and $f_Y(y)$.
3. For the density in Problem 1, compute $\mathbb{P}(Y \le 1/2)$.
4. Let $f(x, y) = 4xy$ on $0 \le x \le 1$, $0 \le y \le 1$. Find the marginal densities. Are X and Y independent?
5. Let $f(x, y) = c(x^2 + y^2)$ on the unit disk. Find c.
6. For independent uniform random variables U, V on $[0, 1]$, define $X = U + V$. Find the density of X by geometric reasoning.
7. Let X, Y have joint density $f(x, y) = x + y$ on the unit square. Compute $\mathbb{E}[X^2]$.
8. Let X, Y have joint density $f(x, y) = x + y$ on the unit square. Compute $\mathrm{Var}(X)$.
9. Let X, Y be independent standard normal random variables. Use polar coordinates to evaluate

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)/2}\, dx\, dy.$$

10. Suppose X has density $f_X(x) = 2x$ on $0 \le x \le 1$, and $Y = X^2$. Find the density of Y.

Likelihood and approximation

1. For Bernoulli data with k successes in n trials, the log-likelihood is

$$\ell(p) = k \log p + (n - k) \log(1 - p).$$

   Find the maximum likelihood estimate of p.

2. Compute the second derivative of the Bernoulli log-likelihood and evaluate it at $\hat{p} = k/n$.
3. For normal data with known variance σ^2, derive the maximum likelihood estimate of μ.
4. For normal data with unknown mean and variance, explain why the log-likelihood is a function of two variables.
5. Let $g(a, b) = a/b$. Compute the linear approximation of g at $(2, 1)$.
6. Let $\Sigma = \begin{bmatrix} 4 & 1 \\ 1 & 2 \end{bmatrix}$. For $g(a, b) = a/b$, approximate the variance of $g(\hat{a}, \hat{b})$ at $(a, b) = (2, 1)$ using the delta method.
7. Let $\ell(\theta_1, \theta_2) = -(\theta_1 - 1)^2 - 4(\theta_2 + 2)^2$. Find the maximum and the Hessian.
8. Explain how the contours of the function in Problem 7 describe uncertainty.

Python exercises

1. Numerically verify that $f(x, y) = x + y$ integrates to 1 on the unit square.
2. Use Monte Carlo simulation to estimate $\mathbb{P}(X + Y \le 1)$ for the density $f(x, y) = x + y$ on the unit square. Compare with the exact answer.
3. Write code to plot the bivariate normal density for several covariance matrices.

4. Simulate uniform points in the unit disk using both $R = U$ and $R = \sqrt{U}$. Compare the radial histograms.
5. Generate normal data and plot the log-likelihood contours for (μ, σ).
6. Simulate the delta method example for different covariance matrices.
7. Estimate $\int_0^1 \int_0^1 \sin(xy)\, dy\, dx$ by Monte Carlo simulation.
8. Use finite differences to approximate the gradient and Hessian of a two-parameter log-likelihood.

24.17 Chapter summary

This chapter showed how the main ideas of multivariable calculus become tools for probability and statistics. Densities are functions integrated over regions. Expectations are weighted integrals. Marginalization is integration over unused variables. Change of variables requires Jacobians. Likelihood functions are multivariable functions on parameter space, and their maxima are studied using gradients and Hessians. Taylor approximation explains why many estimators are approximately normal and why curvature measures uncertainty. These ideas prepare us for statistical theory, Bayesian inference, data science, and machine learning.